- For Individuals
- For Businesses
- For Universities
- For Governments

Explore
Degrees
What do you want to learn?

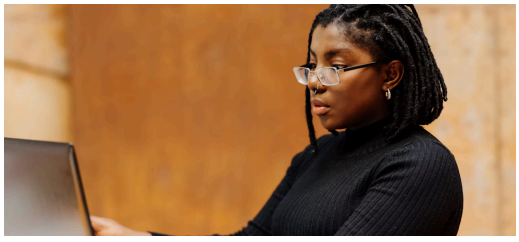
Log In
Join for Free

1.
2. Data
3. Data Science
4. Data Architect vs. Data Engineer: How to Choose the Right Career in Data

Data Architect vs. Data Engineer: How to Choose the Right Career in Data

Written by Coursera Staff • Updated on Feb 24, 2026
Share

Explore how the data architect and data engineer roles compare in terms of their educational requirements, skills, earning potential, and career trajectories to decide which one is best for you.



Key takeaways

Data architects provide the blueprint for an organization's data infrastructure, which data engineers use to build the enterprise's data architecture.

- Since it's a senior-level role, data architects, who earn a median salary of $178,000 annually, tend to earn higher salaries than data engineers, who earn a median salary of $132,000 annually [1,2].
- A data architect's skills focus on data modeling and database design, while a data engineer typically needs to understand how to organize and build data systems and data pipelines.

- You can become a data architect by starting out your career as a data engineer and accumulating several years of experience.

Discover how data architect and data engineer roles compare based on their job responsibilities, required education and skills, and earning potential. If you're ready to start building expertise in data engineering, enroll in the IBM Data Engineering Professional Certificate. In as little as six months, you'll have the opportunity to gain experience with ETL processes, data warehousing, database administration, and other concepts that can benefit your career as a data engineer or even a data architect. Upon completion, you'll have earned a career certificate for your resume.



professional certificate

IBM Data Engineering

Prepare for a career as a Data Engineer. Build job-ready skills – and must-have AI skills – for an in-demand career. Earn a credential from IBM. No prior experience required.

4.6

(62,729 ratings)

189,231 already enrolled

Beginner level

Learn More

Average time: 6 month(s)

Learn at your own pace

Skills you'll build:

Data Analysis, Database Administration, Package and Software Management, Generative AI, Python Programming, Apache Spark, Database Architecture and Administration, Data Store, Professional Networking, Web Scraping, NoSQL, Data Warehousing, IBM Cognos Analytics, Bash (Scripting Language), Data Science, Database Design, Data Pipelines, SQL, Extract, Transform, Load, Apache Hadoop, MongoDB, Apache Cassandra, Query Languages, Database Theory, Scalability, Database Management, Distributed Computing, Databases, Data Manipulation, Interviewing Skills, Linux Commands, Shell Script, File Management, General Networking, Linux, Unix Commands, File Systems, Command-Line Interface, IT Automation, grep, Linux Servers, Unix, Operating Systems, Unix Shell, File I/O, Scripting, Relational Databases, PostgreSQL, MySQL, Data Integrity, Database Management Systems, Data Import/Export, IBM DB2, Data Modeling, User Accounts, Performance Tuning, Disaster Recovery, Identity and Access Management, Data Maintenance, Role-Based Access Control (RBAC), Data Storage Technologies, Database Software, Jupyter, Data Access, Transaction

Processing, Stored Procedure, Data Mart, Data Lakes, Star Schema, Snowflake Schema, Data Quality, Data Validation, Data Architecture, Data Integration, Database Systems, Data Cleansing, Apache Airflow, Big Data, Dashboard Creation, Analytics, Dashboard, Business Intelligence, Model Evaluation, Machine Learning, Unsupervised Learning, Regression Analysis, Supervised Learning, Applied Machine Learning, Model Deployment, Data Processing, Predictive Modeling, Model Training, Classification Algorithms, NumPy, Data Security, Data Governance, Data Storage, Data Synthesis, Data Mining, Data Ethics, Convolutional Neural Networks, Responsible AI, Data Infrastructure, Data Transformation, Application Programming Interface (API), Integrated Development Environments, Style Guides, Unit Testing, Apache Kafka, Data Visualization Software, Looker (Software), Interactive Data Visualization, Data Visualization, Business Analytics, Business Intelligence Software, Debugging, Apache Hive, Application Deployment, Kubernetes, Docker (Software), Development Environment, IBM Cloud

Data architect vs. data engineer: Key responsibilities and daily tasks

A data architect designs data management and storage solutions for an organization, while data engineers implement a data architect's plan by building the systems that facilitate efficient data flow through the data pipeline. A data architect's role is more strategic, focusing on the optimal storage and access of large amounts of data for a business. In contrast, a data engineer's role is on the operational and practical side, constructing and maintaining the infrastructure specified in the data architect's blueprint.

Other responsibilities of data architects include:

- Translating business requirements into integrations, databases, data streams, and data warehouses
- Defining the framework for data architecture and governance standards to ensure data consistency and compliance
- Collaborating with executives, stakeholders, and other departments to decide on the best data management strategy
- Developing patterns for data engineers and other professionals to follow to improve data analytics and systems

Meanwhile, other key responsibilities of data engineers include:

- Developing and managing extract, transform, load (ETL) pipelines to integrate massive volumes of data from multiple sources
- Developing, maintaining, and monitoring databases, data warehouses, and integration tools
- Implementing automation, algorithms, and custom scripts for data analysis, processing, and management
- Monitoring and resolving data pipeline bottlenecks, bugs, or downtimes

Educational requirements and certifications for data engineers vs. data architects

Both data engineers and data architects may begin with a bachelor's degree in computer science, information technology, software engineering, or data science. Data architects may focus on coursework involving data modeling, enterprise architecture, systems development, and information management, while data engineers often take courses in algebra and statistics, database management, computer programming, and system analysis.

Advanced education

Pursuing higher education, like a master's degree, is often required for data architect roles to develop a more comprehensive understanding of data environments and design effective data solutions that meet business needs. Data architects may pursue a master's degree in information systems, big data analytics, data science, and systems engineering, or even business information systems, data or business management, or business architecture to gain business-focused strategic planning skills.

On the other hand, data engineers may choose to reinforce their technical and data analysis skills by pursuing a master's degree in a computer science or data science-related field that includes coursework in cloud computing, big data, and machine learning.

Certifications

You can find popular data engineer certifications that validate skills in areas such as cloud platforms, tools and processes, or fundamental data engineering concepts. Consider the following certifications for data engineers as well as their focus areas:

- **Cloud platforms:** Google Cloud Professional Data Engineer, AWS Certified Data Engineer - Associate
- **Tools and processes:** Databricks Certified Data Engineer Professional, SnowPro Core**,** Cloudera Certified Professional (CCP) Data Engineer
- **Foundational concepts:** Data Science Council of America (DASCA) Associate Big Data Engineer

Data architect certifications typically test your ability to design data and cloud solutions or validate your understanding of data management and enterprise architecture concepts. If you're interested in validating your skills as a data architect, one of the following certifications may be a good option for you:

- **Data and cloud solutions design:** Microsoft Certified: Azure Solutions Architect Expert, AWS Certified Solutions Architect - Associate
- **Data management and enterprise architecture:** DAMA International Certified Data Management Professionals, TOGAF certifications

Can a data engineer become a data architect?

Yes, many data architects start off in data engineer roles and work towards a data architect role by building years of experience in data modeling, data management, systems design, database architecture, etc.



Skills required for a data architect vs. a data engineer

A data architect's skills focus on data modeling and database design, while a data engineer needs to understand how to organize and build data systems and data pipelines. Data architects and data engineers may have an overlapping skill set in data technologies, but how they use these in data management may differ. For example, both data architects and data engineers need to understand cloud platforms such as AWS, Google Cloud, or Microsoft Azure. However, as a data engineer, you will need to know how to leverage cloud technologies to build scalable data systems, while data architects need this skill to plan for expansive data reservoir architectures.

Both data architects and data engineers need to understand databases, data lakes, data warehouses, and ETL processes. In some cases, data architects will have the same skills as data engineers but with more specialized knowledge. For example, data architects need to have an in-depth understanding of data security protocols and governance policies, while data engineers need to know how to follow security procedures to ensure data safety. While a strong command of programming languages like Python, SQL, Java, and Scala is crucial for data engineers to write code for data transformation and pipeline automation, programming skills may not always be mandatory for data architects, although some understanding may be necessary to develop effective data solutions.

In terms of workplace skills, data architects need strategic thinking to plan data architectures that align with business objectives and effective communication to convey technical concepts to nontechnical stakeholders. For data engineers, problem-solving is a key skill for effectively identifying and fixing issues and improving system performance.

IBM

professional certificate

IBM Data Architecture

Launch your career as a data architect. Learn in-demand skills like data engineering, database management, and data architecture in less than 5 months.

4.6

(7,666 ratings)

19,349 already enrolled

Beginner level

Learn More

Average time: 4 month(s)

Learn at your own pace

Skills you'll build:

Law, Regulation, and Compliance, Data Pipelines, Apache Hadoop, Package and Software Management, Data Warehousing, Database Architecture and Administration, Database Design, Data Migration, Database Administration, NoSQL, Incident Response, Data Security, Relational Databases, Data Store, Bash (Scripting Language), Data Architecture, Extract, Transform, Load, SQL, Database Management, Linux Commands, MongoDB, Apache Cassandra, Query Languages, Database Theory, Scalability, Distributed Computing, Databases, Data Manipulation, Data Governance, Information Privacy, Data Management, Regulation and Legal Compliance, Risk Management, Regulatory Compliance, Encryption, Compliance Management, Cryptography, Governance Risk Management and Compliance, Data Infrastructure, Cybersecurity, Security Controls, Computer Security Incident Management, Risk Mitigation, Data Quality, Data Modeling, Data Processing, Data Storage, Software Architecture, Emerging Technologies, Shell Script, File Management, General Networking, Linux, Unix Commands, File Systems, Command-Line Interface, IT Automation, grep, Linux Servers, Unix, Operating Systems, Unix Shell, File I/O, Scripting, PostgreSQL, MySQL, Data Integrity, Database Management Systems, Data Import/Export, IBM DB2, User Accounts, Performance Tuning, Disaster Recovery, Identity and Access Management, Data Maintenance, Role-Based Access Control (RBAC), Data Storage Technologies, Database Software, Data Integration, Data Mart, Data Lakes, Star Schema, Snowflake Schema, Data Validation, Database Systems, Data Cleansing, Cloud Storage, Data Loss Prevention, Interoperability, Data Centers, Big Data, Apache Spark, Data Science, Apache Airflow, Apache Kafka, Data Transformation, Stored Procedure, Transaction Processing, Data Analysis, Microsoft SQL Servers, Debugging, Apache Hive, Application Deployment, Kubernetes, Docker (Software), Development Environment, IBM Cloud

Data engineer vs. data architect salary

Since a data architect is a more senior role than a data engineer, they tend to earn higher salaries due to their expertise in strategic planning and ensuring compliance. Though data engineers also tend to earn competitive salaries, which may vary by experience and company. The table below compares the median total pay for data architects and data engineers in the US, as well as how their salaries change with years of experience.

The salary information below is the median total pay from Glassdoor as of February 2026. These figures include both base salary and additional pay, which may represent profit-sharing, commissions, bonuses, or other forms of compensation.

	Data architect [1]	Data engineer [2]
Median total pay	$178,000	$132,000
1–3 years of experience	$134,000	$116,000
4–6 years of experience	$147,000	$134,000
7–9 years of experience	$162,000	$146,000

What is the difference between data architecture and data engineering growth opportunities and career paths?

Both data architects and data engineers will experience significant demand over the next few years. Global data traffic doubled between 2020 and 2025, and is expected to continue growing at this fast pace [3]. With the rapid increase in data volume, companies will likely continue to need professionals, such as data architects and data engineers, who can plan how to organize their data and build the infrastructure for their data analytics.

As a data engineer, you may specialize in certain fields to work as a big data engineer, cloud data engineer, data integration engineer, or AI/ML engineer. The typical career path of a data engineer is as follows:

- **Entry-level:** You might start your career as a junior data engineer or data analyst, gaining skills in pipeline construction, ETL processes, and data manipulation.
- **Mid-level:** After gaining two to five years of experience, you may be ready to move into a mid-level position like data architect, data scientist, solutions architect, or ETL developer. Your role at this point can involve real-time data processing, working with complex pipelines, mastering cloud computing and data architectures, and developing tailored business solutions.
- **Senior-level:** Once you've accrued five to eight years of experience as a data engineer, more senior-level roles like senior data engineer or data engineering manager may become available to you. In these roles, you might lead teams of data professionals and develop an organization's data strategy.

While data engineering is one pathway to a data architecture role, several other jobs can help you transition into a data architecture role. Discover the general pathway for data architects below.

- **Entry-level:** You can build your experience for a data architect job through positions like data modeler, data warehouse developer, SQL developer, data engineer, or other roles that focus on enterprise architecture.
- **Mid-level:** Once you've gained three to seven years of experience, you can transition to a mid-level role like solutions architect or database administrator. After you've accumulated around nine years of experience, you can explore a data architect role where you may lead the data strategy of an enterprise and ensure data architectures support the organization's operational and analytical goals.
- **Senior-level:** With over 10 years of experience, you can progress to roles like senior data architect, chief data officer, enterprise project manager, or information technology director, overseeing multiple

data projects, leading teams of data architects, and spearheading an organization's digital transformation efforts.

Data scientist vs. data engineer vs. data architect

The data scientist, data engineer, and data architect roles are all interlinked and form an important part of a data science team. The data architect designs and manages a company's data architecture and provides a framework for data storage and management. The data engineer uses this framework to develop and maintain an accessible and well-structured data infrastructure and data pipeline. Data scientists use the data stored in the infrastructure following the processes developed by the data engineer to analyze and extract meaning from the data.



Read more: Data Engineer vs. Data Scientist: What's the Difference?

Learn smarter with resources built for you

If you're looking for more ways to grow, subscribe to our Career Chat newsletter on LinkedIn for industry insights, career tips, and skill-building resources. Or if you want to learn more about the field of data science or what careers and skills to expect, check out these free resources:

- **Take the quiz:** Data Science Career Quiz: Is It Right for You? Find Your Role
- **Watch on YouTube:** 7 Must-Have Skills for Becoming a Data Science Rockstar
- **Hear from an expert in the industry:** 6 Questions with an IBM Data Scientist and AI Engineer

Whether you want to develop a new skill, get comfortable with an in-demand technology, or advance your abilities, keep growing with a Coursera Plus subscription. You'll get access to over 10,000 flexible courses.

Build job-ready skills with Coursera Plus

Start 7-day free trial

Article sources

1.

Glassdoor. “Data Architect Salaries, https://www.glassdoor.com/Salaries/data-architect-salary-SRCH_KO0,14.htm/.” Accessed February 12, 2026.

View all sources

Updated on Feb 24, 2026
Share
Written by:

Coursera Staff

Editorial Team

Coursera’s editorial team is comprised of highly experienced professional editors, writers, and fact...

This content has been made available for informational purposes only. Learners are advised to conduct additional research to ensure that courses and other credentials pursued meet their personal, professional, and financial goals.

Coursera Footer

Skills

- Accounting
- Artificial Intelligence (AI)
- Cybersecurity
- Data Analytics
- Digital Marketing
- Human Resources (HR)
- Microsoft Excel
- Project Management
- Python
- SQL

Professional Certificates

- Google AI Certificate
- Google Cybersecurity Certificate
- Google Data Analytics Certificate
- Google IT Support Certificate
- Google Project Management Certificate
- Google UX Design Certificate
- IBM AI Engineering Certificate
- IBM AI Product Manager Certificate
- IBM Data Science Certificate
- Intuit Academy Bookkeeping Certificate

Courses & Specializations

- AI Essentials Specialization
- AI For Business Specialization
- AI For Everyone Course
- AI in Healthcare Specialization
- Deep Learning Specialization
- Excel Skills for Business Specialization
- Financial Markets Course
- Machine Learning Specialization
- Prompt Engineering for ChatGPT Course
- Python for Everybody Specialization

Career Resources

- Career Aptitude Test
- CAPM Certification Requirements
- CompTIA A+ Certification Requirements
- CompTIA Security+ Certification Requirements

- Essential IT Certifications
- High-Income Skills to Learn
- How to Learn Artificial Intelligence
- PMP Certification Requirements
- Popular Cybersecurity Certifications
- Share your Coursera learning story

Coursera

- About
- Leadership
- Careers
- Catalog
- Coursera Plus
- Professional Certificates
- Degrees
- For Enterprise
- For Government
- For Campus
- Become a Partner
- Udemy

Community

- Learners
- Partners
- Beta Testers
- Free Courses
- Blog
- The Coursera Podcast
- Tech Blog
- Social Impact

More

- Press
- Investors
- Terms
- Privacy
- Help
- Accessibility
- Contact
- Articles

- Directory
- Affiliates
- Modern Slavery Statement
- Cookies Preference Center





© 2026 Coursera Inc. All rights reserved.











